Ontology as a Hyperparameter: Task-Aware Knowledge Graph Construction Using LLMs

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Abstract

This project investigates the construction of task-aware knowledge graphs (KGs) from text, treating graph structure as an optimization target rather than a fixed representation. Existing frameworks such as Neo4j-based pipelines and LLM-driven tools (e.g., LlamaIndex) focus on constructing large, general-purpose graphs without incorporating feedback from downstream tasks, despite evidence from evaluation frameworks (e.g., KGrEaT, GEval, KGBench) that KG utility is task-dependent. The research addresses this gap by proposing a feedback-driven approach in which ontology is treated as a tunable component influencing predictive performance. The study uses document-based datasets linking text to measurable outcomes, including financial news (FNSPID) paired with stock price movements. LLM agents are employed for ontology evolution and triple extraction, enabling scalable generation of multiple KG variants. Graph-derived features are then computed over temporal windows and used in predictive models, whose performance is fed back to the agents to refine ontology design, closing the loop between graph construction and downstream task. Preliminary results indicate that graph structure strongly influences predictive outcomes, with adaptive ontology updates leading to significant improvements.

Contents

1 Introduction

Representing knowledge in structured forms that intelligent systems can query and reason over is a foundational challenge of artificial intelligence. Knowledge graphs (KGs) have become the attractive medium for this purpose, encoding entities, relationships, and semantic descriptions extracted from text to support applications from question answering and recommendation to predictive modelling Ji et al. (2021). The ability to construct KGs from unstructured text at scale has been substantially advanced by large language models (LLMs), which enable flexible extraction of entities and relations without hand-crafted rules Mo et al. (2025). Despite this progress, a fundamental tension persists in KG construction: ontology and graph structure are typically specified *a priori* and evaluated independently of the tasks for which the graph is ultimately used.

Existing KG construction tools — from manual ontology engineering to LLM-driven frameworks such as LlamaIndex and LangChain — treat the ontology as a fixed design choice established before construction begins and unchanged throughout. Evaluation of the resulting graph is typically conducted through structural quality metrics: completeness, consistency, and redundancy Paulheim (2017). Paulheim (2024) draw a sharp distinction between this intrinsic mode of evaluation and task-based evaluation — assessing a KG by its utility on a concrete downstream application. Frameworks such as KGrEaT Heist et al. (2023) and KGBench Bloem et al. (2021) try to formalise the approach to such evaluations. As Heist et al. (2023) observe: *"Performance indicators may be used to make an informed decision when picking a KG for a given task... and to compare different versions of a KG during its life cycle."* Yet despite this insight, no existing work closes the loop at construction time: no lightweight, node-level predictor has been used to directly steer KG ontology evolution during the building process itself

This paper addresses the gap between graph construction and downstream task performance by introducing a feedback-driven framework in which the KG ontology is treated as a tunable hyperparameter. Rather than fixing the schema in advance, we propose a closed-loop system in which: (i) an Ontology Agent proposes candidate ontology configurations; (ii) a KG Builder Agent constructs knowledge graphs from text under each configuration; (iii) graph-derived features are computed over temporal subgraphs and fed into a lightweight downstream predictor; and (iv) predictor performance — measured by AUC and F1-score — is returned as a feedback signal to guide the next ontology revision. This loop repeats until performance converges. Figure **??** illustrates the overall system.

We instantiate this framework on the task of next-day stock price movement prediction using the FNSPID financial news dataset Dong et al. (2024), a corpus of financial news articles linked to NASDAQ price time series. This domain offers a clear and measurable downstream objective, dense entity interactions, and a natural temporal structure that motivates subgraph-based feature extraction. The framework is designed to be domainagnostic and is extensible to other settings, such as biomedical literature linked to adverse drug reactions.

Contributions. This paper makes the following contributions:

1. We propose a feedback-driven KG construction framework that treats ontology as a hyperparameter optimized for downstream task performance, closing the loop between graph construction and prediction.

2. We introduce a structured feature representation for temporal KG subgraphs — comprising node type histograms, relation type histograms, metapath counts, temporal novelty statistics, and topology features — that supports lightweight node-level classification without requiring graph neural networks.

3. We provide empirical evidence on the FNSPID dataset that task-aware ontology adaptation improves predictive accuracy, enables more compact graph representations without sacrificing performance, and produces more informative graph structures over successive iterations.

Paper organisation. Section 2 reviews related work on KG construction, task-based KG evaluation, and task-aware graph learning. Section 3 states the research hypotheses formally. Section 4 describes the proposed framework in detail. Section 5 presents the experimental setup, and Section 6 reports results. Section 7 discusses findings and limitations, and Section 8 concludes.

2 Related Work

Knowledge graphs have become a central representation for structured knowledge extracted from text. Early work on ontology learning **?** established methods for inducing concept hierarchies and relations from corpora using linguistic patterns and statistical co-occurrence. Hogan et al. (2021) provide a comprehensive overview of KG construction, population, and enrichment approaches spanning rule-based systems, distant supervision, and embedding-based methods.

The advent of large language models has substantially lowered the barrier to KG construction at scale. Tools such as LlamaIndex **?**, LangChain **?**, and the Neo4j LLM Knowledge Graph Builder **?** expose pipelines in which an LLM is prompted to extract entities and relations from text, optionally constrained by a user-supplied schema. These tools have seen wide adoption but share a fundamental design choice: the ontology — node types, relation types, and constraints — is specified once and remains fixed for the duration of construction.

More recent academic approaches address extraction quality without revisiting the ontology question. Mo et al. (2025) propose KGGen, which extracts KGs from plain text using language models with improved entity disambiguation. Tsang et al. (2025) apply reinforcement learning to optimise KG construction end-to-end, using graph quality signals as reward. Mynarz and Haniková (2023) propose a test-driven approach in which competency questions drive extraction, providing a form of task specification without a quantitative feedback loop. In all these methods, ontology design remains a pre-construction decision rather than an adaptive variable.

The evaluation of knowledge graphs has historically been conducted through intrinsic, structural quality metrics. Paulheim (2017) survey a broad class of such metrics, covering completeness, consistency, and conciseness. More recently, Paulheim (2024) draw an explicit distinction between this intrinsic mode of evaluation and task-based evaluation — assessing a KG by its utility on a concrete downstream application. This distinction is central to the motivation of the present work: a graph may score well on structural criteria while failing to support a specific predictive task.

Several frameworks have operationalised task-based KG evaluation. KGBench Bloem et al. (2021) provides a suite of relational and multimodal machine learning tasks benchmarked on curated KG datasets. GEval Choi and Jung (2020) offers a software framework for evaluating KGs through downstream task performance across multiple dimensions. KGrEaT Heist et al. (2023) enables systematic comparison of KG variants across classification, regression, and recommendation tasks; the framework explicitly supports comparison of "different versions of a KG during its life cycle," precisely the kind of comparison our feedback loop performs automatically. LLM-KG-Bench **?** focuses specifically on the quality of LLM-generated graphs. Together, these frameworks confirm that task utility cannot be inferred from structural properties alone — yet none of them feeds performance signals back into the construction process itself.

The closest prior work to our approach concerns task-aware graph construction. Cucumides and Geerts (2025) propose constructing graphs from tabular and relational data to maximise GNN performance on a target task, treating graph topology as a task-dependent variable optimised through downstream loss. However, their setting involves structured tabular input rather than unstructured text, and the construction is not driven by an evolving ontology.

In the financial domain, Arun et al. (2025) introduce FinReflectKG, an agentic framework for constructing and evaluating financial knowledge graphs. Their system includes an evaluation component that scores the extracted KG, but feedback is qualitative and does not close a quantitative loop between classifier performance and ontology revision. Both works share the recognition that graph structure should be task-conditioned; neither implements an automated, iterative feedback mechanism of the kind we propose.

A complementary body of work addresses how to derive informative features from knowledge graphs for downstream prediction. Metapath-based approaches Sun et al. (2011); Dong et al. (2017) exploit typed paths in heterogeneous graphs to capture relational semantics between node pairs. Relational Graph Convolutional Networks (R-GCN) Schlichtkrull et al. (2018) extend GCNs to multi-relational graphs, learning embeddings that respect relation type. Knowledge graph embedding methods survey a broad class of techniques for representing entities and relations in continuous space Wang et al. (2017); Ji et al. (2021), supporting link prediction and node-level classification.

These methods share the assumption that the knowledge graph is given and fixed; feature engineering, embedding, or message-passing then operates over the resulting structure. In our framework we draw on metapath counting and relation type histograms as components of the feature vector, but the graph from which these features are derived is itself adapted in response to the downstream task. Rather than learning better from a fixed graph, we ask what graph structure produces the most learnable features.

Taken together, the literature reveals a consistent gap. Evaluation frameworks demonstrate that KG utility is task-dependent Heist et al. (2023); Bloem et al. (2021); task-aware construction methods show that graph structure can be conditioned on downstream objectives Cucumides and Geerts (2025); and graph feature methods show how to extract predictive signal from heterogeneous graphs Sun et al. (2011); Schlichtkrull et al. (2018). Yet no existing work combines these strands into a closed feedback loop in which a lightweight node-level predictor directly steers ontology evolution at construction time. The present paper addresses this gap by treating the KG ontology as a hyperparameter and iteratively refining it based on quantitative downstream performance signals.

3 Research Hypotheses

The proposed framework rests on the premise that knowledge graph ontology is not a neutral design choice but an active determinant of downstream predictive performance. If this is true, then iteratively adapting the ontology in response to task feedback should yield measurable improvements — both in accuracy and in the structural properties of the resulting graph. We formalise this premise through three testable hypotheses, each targeting a distinct dimension of the expected benefit: predictive accuracy, graph efficiency, and structural informativeness.

Hypothesis 1. *Task-aware knowledge graph construction improves prediction accuracy in node-level price movement classification compared to static graph construction approaches.*

Ontology configurations that are better aligned with the prediction objective should produce graph-derived features with higher discriminative power. H1 is evaluated by comparing AUC and F1-score of the downstream classifier trained on features extracted from task-aware KG variants against a static baseline in which the ontology is fixed throughout (Section 6).

Hypothesis 2. *Task-aware optimization enables smaller knowledge graphs while maintaining or improving downstream task performance.*

By discarding ontology elements that do not contribute to predictive performance, the feedback loop should converge on more compact representations. H2 is evaluated by tracking the number of nodes and edges per iteration alongside classifier AUC, testing whether the task-aware variant achieves a favourable size–performance tradeoff relative to the unconstrained baseline (Section 6).

Hypothesis 3. *Iterative, task-driven knowledge graph construction produces more informative graph structures and facilitates the identification of effective ontology configurations.*

Successive ontology revisions guided by performance feedback should increase the diversity and predictive relevance of extracted relation types. H3 is evaluated by examining the ontology evolution trace — which relation types are added, retained, or removed across iterations — and by measuring feature entropy and metapath diversity over successive construction steps (Section 6).

4 Methodology

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5 Experimental Setup

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6 Results

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7 Discussion

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8 Conclusion

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